INCH-POUND
MIL-DTL-24643/55G
27 November 2023
SUPERSEDING
MIL-DTL-24643/55F
1 October 2009

DETAIL SPECIFICATION SHEET

# CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 1,000 VOLTS, TYPE LS4NW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

## Construction (Non-Watertight)

| | | |
|---|---|---|
| First | - | Copper conductor, coated or uncoated 8 AWG, class B stranding, ASTM B8. |
| Second | - | Separator may be used at manufacturer's option where required to provide free-stripping insulation. |
| Third | - | Thermoset insulation with 0.045 inch minimum average thickness. Standard identification code shall be applied by method 1 or method 3. If method 1 is used, the insulation color shall be black. |
| Fourth | - | Four conductors cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive. Fillers may be used as necessary to form a firm, well-rounded assembly. |
| Fifth | - | An optional binder. |
| Sixth | - | Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.050 inch. Maximum overall cable diameter 0.740 inch. |

NOTE 1: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

NOTE 2: LS4NW-8 military part number M24643/55-01UN.



Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:17Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Qualification required.

INSPECTION:

**Basic electricals:**

| | |
|---|---|
| Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.) | 0.679 |
| Voltage withstand (volts, root mean square, min.) | |
| Conductor to conductor | 4,500 |
| Insulation resistance (megohms/1,000 feet, min.) | |
| Conductor to conductor | 4,200 |
| Conductor continuity | No failure |

**Group A:**

| | |
|---|---|
| Visual and dimensional | No failure |

**Group B:**

| | |
|---|---|
| Cold bending (cable) | No failure |
| Thermoset proof test (percent, max.) | |
| Insulation | 50 |
| Jacket (when tested at 392 °F [200 °C]) | 50 |
| Physicals (unaged) | |
| Insulation (extruded insulation only) | |
| Tensile strength ($lb/in^2$, min.) | 700 |
| Elongation (percent, min.) | 150 |
| Jacket (cable) | |
| Tensile strength ($lb/in^2$, min.) | 1,300 |
| Elongation (percent, min.) | 160 |
| Tear (lb/in thickness, min.) | 35 |

**Group C:**

| | |
|---|---|
| Physicals (aged) (air oven) | |
| Insulation (extruded insulation only) | |
| Tensile strength (percent of unaged, min.) | 75 |
| Elongation (percent of unaged, min.) | 75 |
| Jacket (cable) | |
| Tensile strength (percent of unaged, min.) | 60 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:17Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| Elongation (percent of unaged, min.) | 60 |
| Permanence of printing (insulation), method 1 only (cycles, min.) | 25 |
| Permanence of printing (jacket) (cycles, min.) | 125 |
| Heat distortion (percent of unaged, max.) | 30 |
| Shrinkage | No failure |

**Group D:**

| | |
|---|---|
| Flame propagation (cable) | No failure |

**Qualification inspection:**

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

| | |
|---|---|
| Cold working (-4±3.6 °F [-20±2 °C]) | No damage |
| Aging and compatibility (cable) (257±9 °F [125±5 °C]) | No failure |
| Abrasion resistance (jacket) (scrapes, min.) | 75 |
| Acid gas equivalent (percent, max.) | |
| Jacket | 2 |
| Fillers | 2 |
| Insulation | 18 |
| Halogen content (percent, max.) | |
| Jacket | 0.2 |
| Fillers | 0.2 |
| Insulation | 0.2 |
| Immersion (jacket) | |
| Tensile strength (percent of unaged, min.) | 50 |
| Elongation (percent of unaged, min.) | 50 |
| Smoke index (max.) | |
| Jacket | 25 |
| Fillers | 45 |
| Insulation | 45 |
| Toxicity index (max.) | |
| Jacket | 5 |
| Fillers | 5 |
| Insulation | 1.5 |
| Durometer (jacket) – type A (hardness, min.) | 80 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:17Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| Weathering (jacket) | No failure |
| Electrical moisture absorption | No failure |

UNIT ORDERING LENGTHS:

| Type and size | Feet (nominal) |
|---|---|
| LS4NW-8 | 1,500 |

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

## CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-071)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:17Z
Check the source to verify that this is the current version before use.